

- 5 Yeast cells are unicellular eukaryotes that respond to the presence and absence of different sugars by switching genes on or off. One example of this is summarised in Fig. 5.1.

If glucose is **present**, a sequence of events occurs.

- Yeast cells metabolise glucose using constitutively expressed enzymes.
- Mig1 transcription factor (**A**) binds to promoter **B**.
- This stops transcription of gene **C**.
- Production of enzyme **D** stops.

If galactose is present and glucose is **absent**, a different sequence of events occurs.

- The Msn2 transcription factor (**E**) binds to promoter **B**.
- This activates transcription of gene **C**.
- Enzyme **D** is produced and helps convert galactose to glucose.

Gene **F** codes for the Mig1 transcription factor, **A**.

Gene **G** codes for the Msn2 transcription factor, **E**.

Fig. 5.1

- (a) (i) With reference to Fig. 5.1, identify **one** letter corresponding to:

a structural gene

a control (regulatory) sequence

a repressor molecule

[3]

- (ii) Explain why enzyme **D** is described as inducible.

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..... [2]

- (b) Scientists have produced genetically engineered yeast cells. The gene coding for Mig1 transcription factor and the marker gene coding for green fluorescent protein (GFP) are transcribed together to produce a single mRNA molecule.

The resulting Mig1 transcription factor proteins contain a GFP region as part of their structure and are called tagged Mig1 molecules. These tagged Mig1 molecules show up as green fluorescent spots when viewed using a microscope with a very high resolution.

An investigation was carried out to compare the distribution of tagged Mig1 molecules in yeast cells, when glucose is absent and when glucose is present.

The results are shown in Table 5.1.

Table 5.1

glucose availability	mean number of tagged Mig1 molecules present		
	cytoplasm	nucleus	total
glucose absent	1156	176	1332
glucose present	580	226	806

- (i) Calculate the percentage of Mig1 molecules in the nucleus when glucose is **present**.

Show your working and write your answer to **two** significant figures.

..... % [2]

- (ii) When glucose is **absent**, 13% of the available Mig1 molecules are present inside the nucleus.

Explain why this figure is different from your answer to (i).

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..... [2]

- (c) Yeast cells are unable to take in and metabolise the disaccharide sugar lactose.

Some strains of yeast have been genetically engineered to overcome this, by inserting two genes from the bacterium *Escherichia coli* into yeast cells.

Name the **two** bacterial genes that have been inserted into the yeast cells.

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..... [2]

[Total: 11]